Diploma Programme

Physics data booklet

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Diploma Programme
Physics data booklet

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Fundamental constants

Quantity	Symbol	Approximate value
Acceleration of free fall (Earth's surface)	g	$9.81\,\mathrm{m\,s^{-2}}$
Gravitational constant	G	$6.67\times10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$
Avogadro's constant	N_A	$6.02\times10^{23}\,\mathrm{mol^{-1}}$
Gas constant	R	$8.31\,\mathrm{J\,K^{-1}\,mol^{-1}}$
Boltzmann's constant	k_B	$1.38\times10^{-23}\,\mathrm{J\,K^{-1}}$
Stefan–Boltzmann constant	σ	$5.67\times10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$
Coulomb constant	k	$8.99\times10^{9}\,\mathrm{N\,m^2\,C^{-2}}$
Permittivity of free space	ε_0	$8.85\times10^{-12}\,\mathrm{C^2\,N^{-1}\,m^{-2}}$
Permeability of free space	μ_0	$4\pi\times10^{-7}\,\mathrm{T\,m\,A^{-1}}$
Speed of light in vacuum	c	$3.00\times10^{8}\,\mathrm{m\,s^{-1}}$
Planck's constant	h	$6.63\times10^{-34}\,\mathrm{J\,s}$
Elementary charge	e	$1.60\times10^{-19}\,\mathrm{C}$
Electron rest mass	m_e	$9.110\times10^{-31}\,\mathrm{kg} = 0.000549\,\mathrm{u} = 0.511\,\mathrm{MeV\,c^{-2}}$
Proton rest mass	m_p	$1.673\times10^{-27}\,\mathrm{kg} = 1.007276\,\mathrm{u} = 938\,\mathrm{MeV\,c^{-2}}$
Neutron rest mass	m_n	$1.675\times10^{-27}\,\mathrm{kg} = 1.008665\,\mathrm{u} = 940\,\mathrm{MeV\,c^{-2}}$
Unified atomic mass unit	u	$1.661\times10^{-27}\,\mathrm{kg} = 931.5\,\mathrm{MeV\,c^{-2}}$
Solar constant	S	$1.36\times10^{3}\,\mathrm{W\,m^{-2}}$
Fermi radius	R_0	$1.20\times10^{-15}\,\mathrm{m}$

Metric (SI) multipliers

Prefix	Abbreviation	Value
peta	P	10^{15}
tera	T	10^{12}
giga	G	10^{9}
mega	M	10^{6}
kilo	k	10^{3}
hecto	h	10^{2}
deca	da	10^{1}
deci	d	10^{-1}
centi	c	10^{-2}
milli	m	10^{-3}
micro	μ	10^{-6}
nano	n	10^{-9}
pico	p	10^{-12}
femto	f	10^{-15}

Unit conversions

1 radian (rad) $\equiv \frac{180^{\circ}}{\pi}$

Temperature (K) = temperature (°C) + 273

1 light year (ly) = 9.46×10^{15} m

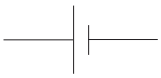
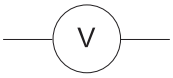
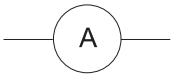
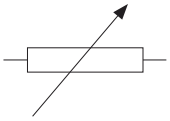
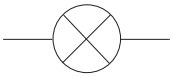
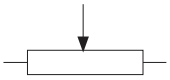
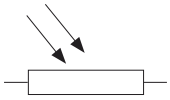
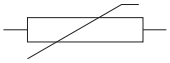
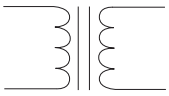
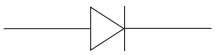
1 parsec (pc) = 3.26 ly

1 astronomical unit (AU) = 1.50×10^{11} m

1 kilowatt-hour (kWh) = 3.60×10^{6} J

$hc = 1.99 \times 10^{-25}$ Jm = 1.24×10^{-6} eV m

Electrical circuit symbols

cell		battery	
ac supply		switch	
voltmeter		ammeter	
resistor		variable resistor	
lamp		potentiometer	
light-dependent resistor (LDR)		thermistor	
transformer		heating element	
diode		capacitor	

Equations—Core

Note: All equations relate to the magnitude of the quantities only. Vector notation has not been used.

Sub-topic 1.2 – Uncertainties and errors

If: $y = a \pm b$ Adding/subtracting quantities: uncertainty in result will be sum of uncertainties of quantities.

then: $\Delta y = \Delta a + \Delta b$

y = Result.

a, b, c = Quantities.

Δ = Uncertainty.

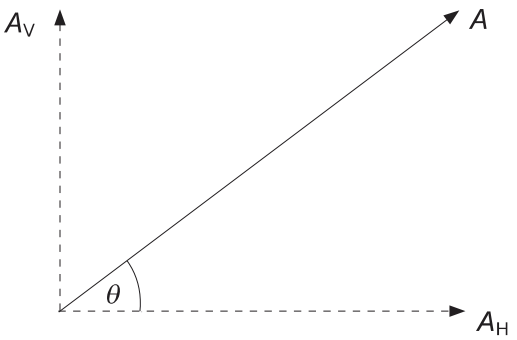
If: $y = \frac{ab}{c}$ Multiplying/dividing quantities: % uncertainties of quantities are added together to obtain % uncertainty in result.

then: $\frac{\Delta y}{y} = \frac{\Delta a}{a} + \frac{\Delta b}{b} + \frac{\Delta c}{c}$

If: $y = a^n$ Powers of quantities: % uncertainty of quantity is multiplied by power to obtain % uncertainty in result.

then: $\frac{\Delta y}{y} = \left| n \frac{\Delta a}{a} \right|$

Sub-topic 1.3 – Vectors and scalars



AH = Horizontal component.

AV = Vertical component.

$A_H = A\cos\theta$ Trigonometric rules of triangles are applied when taking components of vector quantities.

$A_V = A\sin\theta$

F = Resultant force.

m = Mass.

a = Acceleration.

μs = Coefficient of static friction.

μd = " dynamic ".

Ff = Frictional force.

R = Normal reaction force.

Sub-topic 2.1 – Motion

v = Final velocity.

u = Initial velocity.

a = Acceleration ('g' for gravitational).

s = Displacement.

t = Time elapsed.

$v = u + at$

$s = ut + \frac{1}{2}at^2$ Equations applied to uniform motion (known as 'suvat' equations).

$v^2 = u^2 + 2as$

$s = \frac{(v+u)t}{2}$

Sub-topic 2.2 – Forces

$F = ma$ Acceleration due to resultant force (Newton's 2nd law of motion).

$F_f \leq \mu_s R$ Frictional force on a static object.

$F_f = \mu_d R$ Frictional force on a dynamic object.

Sub-topic 2.3 – Work, energy and power

W = Work done.

F = Force.

s = Displacement.

EK = Kinetic energy.

m = Mass.

v = Velocity.

EP = Potential energy.

k = Spring constant

x = Extension.

g = Earth's gravity.

h = Height.

$W = Fs\cos\theta$ Work done.

$E_K = \frac{1}{2}mv^2$ Kinetic energy.

$E_p = \frac{1}{2}k\Delta x^2$ Elastic potential energy (in a spring).

$\Delta E_p = mg\Delta h$ Gravitational potential energy.

$\text{power} = Fv$ Power.

$\text{efficiency} = \frac{\text{useful work out}}{\text{total work in}} = \frac{\text{useful power out}}{\text{total power in}}$

Sub-topic 2.4 – Momentum and impulse

$p = mv$ Momentum.

$F = \frac{\Delta p}{\Delta t}$ Resultant force due to momentum.

$E_K = \frac{p^2}{2m}$ Kinetic energy.

$\text{impulse} = F\Delta t = \Delta p$

p = Momentum.

m = Mass.

v = Velocity.

F = Force.

t = Time.

EK = Kinetic energy.

Q = Energy/heat.

m = Mass.

c = Specific heat capacity.

T = Temperature.

L = Specific latent heat.

Sub-topic 3.1 – Thermal concepts

$Q = mc\Delta T$ Energy/heat given/received in changing an object's temperature.

$Q = mL$ Energy/heat given/received in changing an object's phase.

Sub-topic 3.2 – Modelling a gas

$p = \frac{F}{A}$ Pressure.

$n = \frac{N}{N_A}$ Number of moles of a substance.

$pV = nRT$ Ideal gas law.

$\bar{E}_K = \frac{3}{2}k_B T = \frac{3}{2}\frac{R}{N_A}T$ Average kinetic energy per molecule of a gas.

p = Pressure.

F = Force.

A = Area.

n = Number of moles.

N = Number of atoms.

NA = Avogadro's constant.

V = Volume.

R = Gas constant.

T = Temperature.

EK = Kinetic energy.

kb = Boltzmann's constant.

T = Period.

f = Frequency.

c = Velocity.

f = Frequency.

λ = Wavelength.

I = Intensity.

A = Amplitude.

x = Distance from source.

I0 = Original intensity.

θ = Angle of polarizer.

Sub-topic 4.1 – Oscillations

$T = \frac{1}{f}$ Period (time taken to complete 1 oscillation).

Sub-topic 4.2 – Travelling waves

$c = f\lambda$ Speed of a wave.

Sub-topic 4.3 – Wave characteristics

$I \propto A^2$ Intensity of a wave vs. amplitude.

$I \propto x^{-2}$ Intensity of a wave's radiation at a certain distance from the source.

$I = I_0 \cos^2\theta$ Transmitted intensity of light incident on a polariser (Malus's law).

Sub-topic 4.4 – Wave behaviour

$\frac{n_1}{n_2} = \frac{\sin\theta_2}{\sin\theta_1} = \frac{v_2}{v_1}$ Refraction when a wave crosses a boundary between 2 media (Snell's law).

$s = \frac{\lambda D}{d}$ Fringe spacing in double slit diffraction.

Constructive interference: path difference $= n\lambda$

Destructive interference: path difference $= \left(n + \frac{1}{2}\right)\lambda$

Maxima/minima on screen in double slit diffraction.

n1/n2 = Index of refraction.

θ = Angle of incidence/refraction.

v = Wave velocity.

s = Fringe spacing.

λ = Wavelength.

D = Distance to screen.

d = Slit spacing.

n = Any integer (order of minimum/maximum).

Sub-topic 5.1 – Electric fields

$I = \frac{\Delta q}{\Delta t}$ Current.

$F = k\frac{q_1 q_2}{r^2}$ Force experienced by 2 charges (Coulomb's law).

$k = \frac{1}{4\pi\varepsilon_0}$ Coulomb constant.

$V = \frac{W}{q}$ Potential difference.

$E = \frac{F}{q}$ Electric field strength.

$I = nAvq$ Current in a wire.

I = Current.
q = Charge.
t = Time.
F = Force.
k = Coulomb constant.
r = Separation distance.
ε0 = Permittivity of free space.
V = Potential.
W = Work done.
E = Electric field strength.
n = Number of charges per unit volume.
A = X-sectional area.
v = Drift velocity.

Sub-topic 5.2 – Heating effect of electric currents

Kirchhoff's circuit laws:

$\Sigma V = 0$ (loop)

$\Sigma I = 0$ (junction)

$R = \frac{V}{I}$ Resistance.

$P = VI = I^2R = \frac{V^2}{R}$ Power supplied/dissipated.

$R_{total} = R_1 + R_2 + \ldots$ Total resistance of resistors in series.

$\frac{1}{R_{total}} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$ Total resistance of resistors in parallel.

$\rho = \frac{RA}{L}$ Resistivity of material of a wire.

V = Potential.
I = Current.
R = Resistance.
P = Power.
ρ = Resistivity.
A = X-sectional area.
L = Length.

Sub-topic 5.3 – Electric cells

$\varepsilon = I(R + r)$ Emf of a cell.

ε = Emf.
I = Current.
R = Resistance.
r = Internal resistance.

Sub-topic 5.4 – Magnetic effects of electric currents

$F = qvB\sin\theta$ Force on a charge moving through a magnetic field.

$F = BIL\sin\theta$ Force on a current-carrying wire in a magnetic field.

F = Force.
q = Charge.
v = Velocity of charge.
B = Magnitude of magnetic field.
θ = Angle with field.

Sub-topic 6.1 – Circular motion

$v = \omega r$ Velocity of body travelling in circle.

$a = \frac{v^2}{r} = \frac{4\pi^2 r}{T^2}$ Centripetal acceleration.

$F = \frac{mv^2}{r} = m\omega^2 r$ Centripetal force.

v = Velocity.
ω = Angular velocity.
r = Radius of circle.
a = Acceleration.
T = Period of rotation.
F = Force.
m = Mass.

Sub-topic 6.2 – Newton's law of gravitation

$F = G\frac{Mm}{r^2}$ Force experienced by 2 masses (Newton's law of gravitation).

$g = \frac{F}{m}$ Field strength as experienced by a mass in the field.

$g = G\frac{M}{r^2}$ Field strength at a certain distance from body.

F = Force.
G = Gravitational constant.
M = Mass of body.
m = Mass of body (in a field).
r = Separation distance of bodies.
g = Gravitational field strength

Sub-topic 7.1 – Discrete energy and radioactivity

$E = hf$ Energy of a photon.

E = Energy.

h = Planck's constant.

f = Frequency.

$\lambda = \frac{hc}{E}$ Wavelength of a photon.

λ = Wavelength.

c = Speed of light.

Sub-topic 7.2 – Nuclear reactions

$\Delta E = \Delta mc^2$ Energy released when nucleons are assembled into nucleus.

E = Energy.

m = Mass.

c = Speed of light.

Sub-topic 7.3 – The structure of matter

Charge	Quarks			Baryon number
$\frac{2}{3}e$	u	c	t	$\frac{1}{3}$
$-\frac{1}{3}e$	d	s	b	$\frac{1}{3}$
All quarks have a strangeness number of 0 except the strange quark that has a strangeness number of –1				

e = Elementary charge.

u = Up.

d = Down.

c = Charm.

s = Strange.

t = Top.

b = Bottom.

Charge	Leptons		
–1	e	μ	τ
0	ν_e	ν_μ	ν_τ
All leptons have a lepton number of 1 and antileptons have a lepton number of –1			

e = Electron.

u = Muon.

τ = Tau.

ν = Neutrino.

	Gravitational	Weak	Electromagnetic	Strong
Particles experiencing	All	Quarks, leptons	Charged	Quarks, gluons
Particles mediating	Graviton	W^+, W^-, Z^0	γ	Gluons

Sub-topic 8.1 – Energy sources

$\text{power} = \frac{\text{energy}}{\text{time}}$

$\text{power} = \frac{1}{2}A\rho v^3$ Power available from a wind turbine.

A = Area swept out by turbine blades.

ρ = Air density.

v = Wind speed.

Sub-topic 8.2 – Thermal energy transfer

$P = e\sigma AT^4$ Power radiated by a body.

$\lambda_{max}(\text{metres}) = \frac{2.90 \times 10^{-3}}{T(\text{kelvin})}$ Wavelength at which intensity of radiation is at a maximum.

$I = \frac{\text{power}}{A}$ Intensity of radiation.

$\text{albedo} = \frac{\text{total scattered power}}{\text{total incident power}}$

P = Power.

e = Emissivity.

σ = Stefan-Boltzmann constant.

A = Area.

T = Temperature.

λ = Wavelength.

I = Intensity.

Equations—AHL

Sub-topic 9.1 – Simple harmonic motion

$\omega = \frac{2\pi}{T}$ Angular frequency of oscillation.

$a = -\omega^2 x$ Acceleration of object in SHM.

$x = x_0 \sin \omega t;\ x = x_0 \cos \omega t$ Displacement of object in SHM.

$v = \omega x_0 \cos \omega t;\ v = -\omega x_0 \sin \omega t$ Velocity of object in SHM.

$v = \pm\omega\sqrt{(x_0^2 - x^2)}$ Velocity of object in SHM.

$E_K = \frac{1}{2} m\omega^2 (x_0^2 - x^2)$ Kinetic energy of object in SHM.

$E_T = \frac{1}{2} m\omega^2 x_0^2$ Total energy of object in SHM.

pendulum: $T = 2\pi\sqrt{\frac{l}{g}}$ Period of oscillation of a pendulum in SHM.

mass-spring: $T = 2\pi\sqrt{\frac{m}{k}}$ Period of oscillation of a mass on a spring in SHM.

ω = Angular frequency.

T = Period.

a = Acceleration.

x = Displacement from equilibrium.

x0 = Maximum displacement.

t = Time elapsed.

EK = Kinetic energy.

ET = Total energy.

l = Length of pendulum.

g = Gravitational field strength.

k = Spring constant.

Sub-topic 9.2 – Single-slit diffraction

$\theta = \frac{\lambda}{b}$ Angle at which first minimum occurs in single-slit diffraction.

θ = Angle.

λ = Wavelength.

b = Slit width.

Sub-topic 9.3 – Interference

$n\lambda = d \sin\theta$ Path difference between slits for a diffraction grating (constructive/destructive interference).

Constructive interference: $2dn = \left(m + \frac{1}{2}\right)\lambda$

Destructive interference: $2dn = m\lambda$

Interference patterns for thin-film interference.

n = Any integer (for diffraction grating).

λ = Wavelength.

d = Slit spacing (for diffraction grating).

θ = Angle.

d = Thickness of medium (for TFI).

n = Refractive index of medium (for TFI).

m = Any integer (for TFI).

Sub-topic 9.4 – Resolution

$\theta = 1.22\frac{\lambda}{b}$ First minimum for diffraction in a circular aperture.

$R = \frac{\lambda}{\Delta\lambda} = mN$ Resolvance of a diffraction grating.

θ = Angle.

λ = Wavelength.

b = Slit width/diameter.

R = Resolvance

Δλ = Smallest possible resolvable wavelength difference.

m = Diffraction order.

N = Number of slits illuminated.

Sub-topic 9.5 – Doppler effect

Moving source: $f' = f\left(\frac{v}{v \pm u_s}\right)$

Moving observer: $f' = f\left(\frac{v \pm u_o}{v}\right)$

$\frac{\Delta f}{f} = \frac{\Delta\lambda}{\lambda} \approx \frac{v}{c}$ Doppler effect for light.

f′ = Perceived frequency.

f = Actual frequency.

v = Wave speed.

us = Velocity of source.

uo = Velocity of observer.

λ = Wavelength.

v = Relative speed of observer and source.

c = Speed of light.

W = Work done.

q = Charge.

Ve = Electric potential.

m = Mass.

Vg = Gravitational potential.

Sub-topic 10.1 – Describing fields

$W = q\Delta V_e$

Work done moving a charge between 2 points in a field.

Potential.

$W = m\Delta V_g$

Work done moving a mass between 2 points in a field.

Field strength.

Potential energy.

Force.

Sub-topic 10.2 – Fields at work

$V_g = -\frac{GM}{r}$	$V_e = \frac{kQ}{r}$
$g = -\frac{\Delta V_g}{\Delta r}$	$E = -\frac{\Delta V_e}{\Delta r}$
$E_p = mV_g = -\frac{GMm}{r}$	$E_p = qV_e = \frac{kQq}{r}$
$F_g = \frac{GMm}{r^2}$	$F_e = \frac{kQq}{r^2}$

$v_{esc} = \sqrt{\frac{2GM}{r}}$ Escape velocity of a planet.

$v_{orbit} = \sqrt{\frac{GM}{r}}$ Velocity of a body in circular orbit around another body.

Vg = Gravitational potential.

Ve = Electric potential.

G = Gravitational constant.

k = Coulomb constant.

M = Mass.

Q = Charge.

r = Separation distance.

g = Gravitational field strength.

E = Electric field strength.

Ep = Potential energy.

m = Mass.

q = Charge.

Fg = Gravitational force.

Fe = Electric force.

V(esc) = Escape velocity.

V(orbit) = velocity of orbit.

Sub-topic 11.1 – Electromagnetic induction

$\Phi = BA\cos\theta$ Magnetic flux.

$\varepsilon = -N\frac{\Delta\Phi}{\Delta t}$ Induced emf in a coil.

$\varepsilon = Bvl$ Induced emf in a conductor moving through a field.

$\varepsilon = BvlN$ Induced emf in a coiled wire moving through a field.

Sub-topic 11.2 – Power generation and transmission

$I_{rms} = \frac{I_0}{\sqrt{2}}$ Effective (root mean square) current in an AC generator.

$V_{rms} = \frac{V_0}{\sqrt{2}}$ Effective (root mean square) potential difference in an AC generator.

$R = \frac{V_0}{I_0} = \frac{V_{rms}}{I_{rms}}$ Resistance.

$P_{max} = I_0 V_0$ Maximum power dissipated.

$\bar{P} = \frac{1}{2} I_0 V_0$ Average power dissipated.

$\frac{\varepsilon_p}{\varepsilon_s} = \frac{N_p}{N_s} = \frac{I_s}{I_p}$ Ratios of emf, turns and current in a transformer.

Sub-topic 11.3 – Capacitance

$C = \frac{q}{V}$ Capacitance of a capacitor.

$C_{parallel} = C_1 + C_2 + \ldots$ Capacitance of capacitors in parallel.

$\frac{1}{C_{series}} = \frac{1}{C_1} + \frac{1}{C_2} + \ldots$ Capacitance of capacitors in series.

$C = \varepsilon\frac{A}{d}$ Capacitance of a capacitor.

$E = \frac{1}{2}CV^2$ Energy stored in a capacitor.

$\tau = RC$ Time constant for a circuit.

$q = q_0 e^{-\frac{t}{\tau}}$ Exponential decrease of charge stored for a discharging capacitor.

$I = I_0 e^{-\frac{t}{\tau}}$ Exponential decrease of current for a discharging capacitor.

$V = V_0 e^{-\frac{t}{\tau}}$ Exponential decrease of potential difference for a discharging capacitor.

Φ = Magnetic flux.

B = B = Magnitude of magnetic field.

A = Area of coil.

N = Number of turns.

t = Time elapsed.

v = Speed of wire.

l = Length of wire.

I(rms) = Effective current.

I0 = Maximum current.

V(rms) = Effective pd.

V0 = Maximum pd.

R = Resistance

P(max) = Maximum power dissipated.

P = Power dissipated.

ε = Emf.

N = Number of turns.

p/s = Primary/ secondary.

C = Capacitance.

q = Charge.

V = Potential (difference).

ε = Permittivity of dielectric material.

A = Area of plates.

d = Separation of plates.

E = Energy stored.

τ = Time constant.

R = Resistance.

q0 = Original charge.

t = Time elapsed.

I = Current.

I0 = Initial maximum current.

V0 = Initial maximum potential difference.

Sub-topic 12.1 – The interaction of matter with radiation

$E = hf$ Energy of a photon.

$E_{max} = hf - \Phi$ Kinetic energy of freed electron (photoelectric effect) (= e × stopping voltage).

$E = -\frac{13.6}{n^2}eV$ Quantised energy of electron in the hydrogen atom.

$mvr = \frac{nh}{2\pi}$ Angular momentum of the orbiting electron in the hydrogen atom.

$P(r) = |\psi|^2 \Delta V$ Probability that an electron will be found within a small volume ΔV.

$\Delta x \Delta p \geq \frac{h}{4\pi}$ Uncertainty in momentum and position of a particle (Heisenberg).

$\Delta E \Delta t \geq \frac{h}{4\pi}$ Uncertainty in energy and lifetime of the state of a particle (Heisenberg).

Sub-topic 12.2 – Nuclear physics

$R = R_0 A^{\frac{1}{3}}$ Nuclear radius of an element.

$N = N_0 e^{-\lambda t}$ Number of nuclei left in a radioactive sample.

$A = \lambda N_0 e^{-\lambda t}$ Activity of a radioactive sample.

$\sin\theta \approx \frac{\lambda}{D}$ First minimum of an electron diffraction pattern around a circular object.

E = Energy.

h = Planck's constant.

f = Frequency.

Φ = Work function.

n = State of atom.

m = Mass.

v = Velocity.

r = Radius.

Ψ = Wave function.

V = Volume.

x = Position.

p = Momentum.

t = Time.

R = Nuclear radius.

R0 = Fermi radius (constant).

A = Atomic mass number.

N = Number of nuclei.

N0 = Original number of nuclei.

A = Activity.

λ = Decay constant.

θ = Angle of first minimum.

λ = De Broglie wavelength.

D = Diameter of circular object.

Equations—Options

<table>
<tr><th>Sub-topic A.1 – The beginnings of relativity</th><th>Sub-topic A.2 – Lorentz transformations</th></tr>
<tr><td>$x' = x - vt$

$u' = u - v$</td><td rowspan="2">$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$

$x' = \gamma(x - vt);\ \Delta x' = \gamma(\Delta x - v\Delta t)$

$t' = \gamma\left(t - \frac{vx}{c^2}\right);\ \Delta t' = \gamma\left(\Delta t - \frac{v\Delta x}{c^2}\right)$

$u' = \frac{u - v}{1 - \frac{uv}{c^2}}$

$\Delta t = \gamma \Delta t_0$

$L = \frac{L_0}{\gamma}$

$(ct')^2 - (x')^2 = (ct)^2 - (x)^2$</td></tr>
<tr><th>Sub-topic A.3 – Spacetime diagrams</th></tr>
<tr><td>$\theta = \tan^{-1}\left(\frac{v}{c}\right)$</td><td></td></tr>
<tr><th>Sub-topic A.4 – Relativistic mechanics (HL only)</th><th>Sub-topic A.5 – General relativity (HL only)</th></tr>
<tr><td>$E = \gamma m_0 c^2$

$E_0 = m_0 c^2$

$E_K = (\gamma - 1) m_0 c^2$

$p = \gamma m_0 v$

$E^2 = p^2 c^2 + m_0{}^2 c^4$

$qV = \Delta E_K$</td><td>$\frac{\Delta f}{f} = \frac{g\Delta h}{c^2}$

$R_s = \frac{2GM}{c^2}$

$\Delta t = \frac{\Delta t_0}{\sqrt{1 - \frac{R_s}{r}}}$</td></tr>
</table>

Sub-topic B.1 – Rigid bodies and rotational dynamics	Sub-topic B.2 – Thermodynamics
$\Gamma = Fr\sin\theta$ $I = \sum mr^2$ $\Gamma = I\alpha$ $\omega = 2\pi f$ $\omega_f = \omega_i + \alpha t$ $\omega_f^2 = \omega_i^2 + 2\alpha\theta$ $\theta = \omega_i t + \frac{1}{2}\alpha t^2$ $L = I\omega$ $E_{K_{rot}} = \frac{1}{2}I\omega^2$	$Q = \Delta U + W$ $U = \frac{3}{2}nRT$ $\Delta S = \frac{\Delta Q}{T}$ $pV^{\frac{5}{3}} =$ constant (for monatomic gases) $W = p\Delta V$ $\eta = \frac{\text{useful work done}}{\text{energy input}}$ $\eta_{Carnot} = 1 - \frac{T_{cold}}{T_{hot}}$
Sub-topic B.3 – Fluids and fluid dynamics (HL only)	**Sub-topic B.4 – Forced vibrations and resonance (HL only)**
$B = \rho_f V_f g$ $P = P_0 + \rho_f gd$ $Av =$ constant $\frac{1}{2}\rho v^2 + \rho gz + p =$ constant $F_D = 6\pi\eta rv$ $R = \frac{vr\rho}{\eta}$	$Q = 2\pi\frac{\text{energy stored}}{\text{energy dissipated per cycle}}$ $Q = 2\pi \times \text{resonant frequency} \times \frac{\text{energy stored}}{\text{power loss}}$

Sub-topic C.1 – Introduction to imaging

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$P = \frac{1}{f}$$

$$m = \frac{h_i}{h_o} = -\frac{v}{u}$$

$$M = \frac{\theta_i}{\theta_o}$$

$$M_{\text{near point}} = \frac{D}{f} + 1; M_{\text{infinity}} = \frac{D}{f}$$

Sub-topic C.2 – Imaging instrumentation

$$M = \frac{f_o}{f_e}$$

Sub-topic C.3 – Fibre optics

$$n = \frac{1}{\sin c}$$

$$\text{attenuation} = 10 \log \frac{I}{I_0}$$

Sub-topic C.4 – Medical imaging (HL only)

$$L_I = 10 \log \frac{I_1}{I_0}$$

$$I = I_0 e^{-\mu x}$$

$$\mu x_{\frac{1}{2}} = \ln 2$$

$$Z = \rho c$$

d = Distance from Earth to a star.

p = Parallax angle.

L = Luminosity.

σ = Stefan-Boltzmann constant.

A = Area.

T = Temperature.

b = Apparent brightness.

d = Distance to star.

z = Red shift.

λ(0) =Emitted wavelength.

v = Relative velocity of light source.

c = Speed of light.

R = Cosmic scale factor.

R(0) =

H(0) = Hubble constant.

d = Distance from Earth.

Sub-topic D.1 – Stellar quantities

$$d\,(\text{parsec}) = \frac{1}{p\,(\text{arc-second})}$$ Distance to a star in parsec.

$$L = \sigma A T^4$$ Luminosity of a star.

$$b = \frac{L}{4\pi d^2}$$ Apparent brightness of a star.

Sub-topic D.2 – Stellar characteristics and stellar evolution

$$\lambda_{\max} T = 2.9 \times 10^{-3}\ \text{mK}$$ Relation between wavelength of maximum intensity radiation of a star and its temperature.

$$L \propto M^{3.5}$$ Mass-luminosity relation for main sequence stars.

λ = Wavelength.

T = Temperature.

L = Luminosity.

M = Mass.

Sub-topic D.3 – Cosmology

$$z = \frac{\Delta\lambda}{\lambda_0} \approx \frac{v}{c}$$ Red shift of a star/galaxy moving away from us.

$$z = \frac{R}{R_0} - 1$$ Red shift depending on cosmic scale factor.

$$v = H_0 d$$

$$T \approx \frac{1}{H_0}$$

Sub-topic D.5 – Further cosmology (HL only)

$$v = \sqrt{\frac{4\pi G\rho}{3}}\, r$$

$$\rho_c = \frac{3H^2}{8\pi G}$$